

H.M. Shadman Tabib

+880-1880-198766 | shadmantabib2002@gmail.com | [Portfolio](#)

[Google Scholar](#) [hm-shadman-tabib](#) | [shadmantabib](#)

RESEARCH INTERESTS

Computer Vision & AI for Health: Medical imaging , multimodal/semi-supervised learning, bioimage segmentation, real-time systems.

Machine Learning: Unsupervised detection/segmentation, contextualized ML, LLM-as-a-judge

EDUCATION

- University of Illinois Urbana-Champaign (UIUC)** Started in Fall 2026
First year Ph.D. Student in Computer Science, Siebel School of Computing and Data Science
Admitted to the Ph.D. program in Computer Science, starting Fall 2026. Urbana-Champaign, IL, USA
- Bangladesh University of Engineering and Technology (BUET)** 2022 – 2026
Department of Computer Science and Engineering
Awarded Dean's List Scholarship and University Merit Scholarship multiple times . Dhaka, Bangladesh
CGPA: 3.88 / 4.00

RESEARCH EXPERIENCE

- Carnegie Mellon University (Xu Lab)** Jan 2024 – Present
Remote Research Collaborator Pittsburgh, PA, USA (Remote)
Advisor: [Dr. Min Xu](#)
Worked on unsupervised Cryo-ET segmentation [**Submitted**] and prepared novel particle picking algorithm [**Accepted**].
Conducted denoising for high-resolution biological structure reconstruction.
- University of WisconsinMadison (Adaptive Inference Lab)** Sep 2024 – Sep 2025
Remote Research Collaborator Madison, WI, USA (Remote)
Advisor: [Dr. Ben Lengerich](#)
Co-authored a comprehensive review on contextualized machine learning and interpretability in foundation models.
Contributing to the AdaptInfer framework and contextual awareness in data-driven modeling. [**Completed**]
- Bangladesh University of Engineering and Technology (BUET)** Jan 2023 – Present
Undergraduate Research Assistant Dhaka, Bangladesh
Supervisor: [Prof. Dr. M. Sohel Rahman](#)
Semi-supervised and multimodal breast cancer classification using contextualized vision pipelines for medical imaging [**Accepted**] and connectomics EM segmentation. [**Ongoing**]
Supervisor: [Prof. Dr. Md. Shamsuzzoha Bayzid](#)
Developing RNA task accuracy estimation leveraging multimodal inputs involving self supervised losses, and multimodal model training frameworks for biological connectomics [**Ongoing**]
Supervisor: [Prof. Dr. A.B.M. Alim Al Islam](#)
Designed EEG-based context-aware deep learning systems for single-electrode epilepsy prognosis and real-time braincomputer interface modeling.[**Accepted**]
Supervisor: [Prof. Dr. Ch. Md. Rakin Haider](#)
Developing SpectraSentinel a dual-stream, lightweight real-time drone detection and tracking system combining spectral and spatial fusion cues. [**Submitted**]
Supervisor: [Dr. K.M. Ariful Kabir](#)
Modeling stochastic spatio-temporal epidemiology on multiplex networks; simulating socioeconomic epidemic spread through agent-based models using Mesa.[**Submitted**]

PUBLICATIONS

(Citation count: 17 in Google Scholar)

- Unsupervised Multi-scale Segmentation of Cellular Cryo-electron Tomograms with Stable Diffusion Foundation Model.**
Mostofa Rafid Uddin, **H.M. Shadman Tabib**, Thanh-Huy Nguyen, Kashish Gandhi, Min Xu.
 CVPR 2026 Main Track Accepted (Highlight and Award Candidate)[[pdf](#)] 2026
- Localization of Macromolecules in Crowded Cellular Cryo-electron Tomograms from Extremely Sparse Labels.**
Mostofa Rafid Uddin, **H.M. Shadman Tabib***, Ajmain Yasar Ahmed*, Md Toki Tahmid, Md Zarif Ul Alam, Zachary Freyberg, Min Xu.
 Briefings in Bioinformatics Accepted [[pdf](#)] 2025

- [3] **NEUROSKY-EPI: The First Open Single-Electrode Epilepsy EEG Dataset with Context-Aware Modeling.**
H.M. Shadman Tabib, Md. Hasnaen Adil, Ayesha Rahman, Ahmmad Nur Swapnil, Maoyejatun Hasana, A.B.M. Alim Al Islam.
📄 [NeurIPS 2025 Workshop on Time Series for Health \(TS4H\)](#) Accepted [\[OpenReview\]](#) 2025
- [4] **Contextualized Machine Learning: Towards Adaptive, Interpretable, and Generalizable AI Systems.**
Ben Lengerich, Caleb Ellington, Sazan Mahbub H.M. Shadman Tabib, et al.
📄 [UWMadison Research Project \[GitHub\]](#) 2025
- [5] **End-to-End Bangla AI for Solving Math Olympiad Problem Benchmark.**
H.M. Shadman Tabib, Jaber Ahmed Deedar.
📄 [International Journal of Natural Language Computing \(IJNLC\)](#) Accepted [\[pdf\]](#) 2024
- [6] **DeCoDrift: Stabilizing Decoder Coupling in Closed-Loop Foundation Segmentation.**
H.M. Shadman Tabib, Md. Shamsuzzoha Bayzid, M. Sohel Rahman.
📄 [arXiv \[pdf\]](#) 2026
- [7] **Spectral Forensics of Diffusion Attention Graphs for Copy-Move Forgery Detection.**
H. M. Shadman Tabib, Tasriad Ahmed Tias, Nafis Tahmid.
📄 [arXiv \[pdf\]](#) 2026
- [8] **Breast Cancer Detection Using Semi-Supervised Learning: Multimodal Data Integration and Comparative Demographic Analysis.**
H.M. Shadman Tabib, Mst. Fahmida Sultana Naznin, Sadatul Islam Sadi, Md. Hasnaen Adil, Zarin Tasnim Oishi, M. Sohel Rahman.
📄 [NSyS 2024 \(11th Intl. Conf. on Networking, Systems, and Security\)](#) - Accepted as Poster [\[pdf\]](#) 2024
- [9] **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization.**
H.M. Shadman Tabib et al.
📄 [arXiv \[pdf\]](#) 2026
- [10] **Study on Locomotive Epidemic Dynamics in a Stochastic Spatio-Temporal Simulation Model on a Multiplex Network.**
H.M. Shadman Tabib*, Jaber Ahmed Deedar*, K.M. Ariful Kabir.
📄 [arXiv \[pdf\]](#) 2025
- [11] **SpectraSentinel: Lightweight Dual-Stream Real-Time Drone Detection, Tracking and Payload Identification.**
Shahriar Kabir, Istiak Ahmmmed Rifti, H.M. Shadman Tabib, Mushfiqur Rahman, Sadatul Islam Sadi, Hasnaen Adil, Ahmed Mahir Sultan Rumi, Ch. Md. Rakin Haider.
📄 [arXiv \[pdf\]](#) 2025
- [12] **Exploring Livelihood Dynamics and Policy Interventions in Mangrove Social-Ecological Systems with Agent-Based Modeling: A Mesa Framework Approach.**
Anik Saha, H.M. Shadman Tabib, M. Sohel Rahman.
📄 [arXiv \[PDF\]](#) 2024
- [13] **Toward Trustworthy Difficulty Assessments: Large Language Models as Judges in Programming and Synthetic Tasks.**
H.M. Shadman Tabib, Jaber Ahmed Deedar.
📄 [arXiv \[pdf\]](#) 2025

WORK EXPERIENCE

- 🏛️ **National University of Singapore (NUS)** Jun 2026 – July 2026
Singapore
Visiting Scholar, SRI Program
Working as an on-site research intern under the SRI Program, focusing on health informatics. I worked on rigorous preprocessing of multi-modal health data using frontier Language Models
- 🏢 **Bangladesh University of Engineering and Technology (BUET)** Feb 2026 – July 2026
Dhaka, Bangladesh
Machine Learning Engineering Expert
Working part-time on Education AI, with a focus on developing question-generation modules based on frontier generative models. Collaborating on an adaptive question generation pipeline for personalized and curriculum-aligned assessment.
- 🏢 **Research & Innovation Centre (RIC), Bangladesh University of Engineering and Technology (BUET)** Feb 2025 – Nov 2025
Dhaka, Bangladesh
Part-Time Research Engineer
Project: Identification of Dengue Breeding Sites through Object Recognition (HE-01-244).
Contributed to dengue vector surveillance using deep learning-based object recognition and image analysis for public health applications.

AWARDS & RECOGNITIONS

- **Global Champion Johns Hopkins Healthcare Design Competition (Team NeoScreenix)** 2025
Official Competition Page; National Daily News Coverage: [Daily Star](#), [TBS News](#), [UNB News](#); TV Coverage: [YouTube](#).
Developed a smart AI system for early breast cancer screening integrating multimodal imaging and explainable deep learning. Won 1st place among 440+ global teams and 200+ universities worldwide in the Digital Health category, representing Bangladesh University of Engineering and Technology (BUET).
- **Global 2nd Runner-up IEEE SPS Video & Image Processing (VIP) Cup** 2025
International competition on video & image processing.
Designed a vision-based drone detection and payload identification system using RGBIR fusion models. Implemented real-time detection and tracking pipelines optimized for UAV surveillance, achieving top-tier accuracy and inference speed among worldwide university teams.
- **Global Finalist Bradford Quantum Hackathon (Team NeuroCat)** 2025
International hackathon on quantum computing and clinical trial optimization, organized by Quantinuum and the University of Bradford.
Finalist in the Clinical Trial Optimization track hosted on the Aqora platform using the Mayo Clinic dataset. Developed a hybrid quantumclassical optimization pipeline featuring QUBO formulation, D-Wave quantum annealing, hardware-aware QAOA on IBM NISQ devices, and warm-start hybrid solver strategies for scalability.
- **Datathons** 2024 – 2025
KUET 2025 (Top 5%); BUET DL Sprint 2024 (Top 15%); CZI Kaggle 2024 (Top 43%); SUST DL Enigma 2024 (Top 34%)
- **National Top 7 - Code Refactoring Contest (Code Catalyst 2024)** 2024
National-level code refactoring and software quality competition held at IUT ICT Fest 2024. Focused on identifying and resolving code smells, improving maintainability, readability, and structural design. [[GitHub Repository](#)].
- **Deans & University Merit Scholarships BUET** 2022 – Present
Sustained academic excellence (CGPA > 3.75 over three consecutive years) and topped in the department
- **Speaker Recognition IEEE SPS & IEEE CS, CUET** 2025
Certificate; Invited speaker for the webinar Unlocking IEEE SP Cup & VIP Cup: A Beginner's Guide.
- **Champion Bangladesh Biology Olympiad (BdBO)** 2019-20
- **Second Runner-up Bangladesh Mathematical Olympiad (BdMO)** 2019–2020
- **Winner Bangladesh Physics Olympiad (BdPhO)** 2019
- **4th Place & Extended National Camper Bangladesh Chemistry Olympiad (BdChO)** 2020
Ranked 4th nationally and selected for extended national camp to represent Bangladesh at IChO 2020.

ENTREPRENEURSHIP & LEADERSHIP

- **Co-Founder & Head of Research — PinkLifeLine** Jan 2025 — Present
 - Health-tech startup from NeoScreenix (JHU 2025 Global Champion); funded by Bangladesh National ICT Division.
 - **Collaborations:** Partnered with [Sustainlaunch Labs](#) (global innovation accelerator) and [Herwill](#) (USA based global women's empowerment & digital inclusion).
 - Built and leads a **50+ medical professionals' network**; leads ML clinical data pipelines & screening workflows.
- **Lead Organizer & Problem Setter — BUET CSE Fest Deep Learning Sprint 2025** Nov 2025 — Jan 2026
 - Coordinated largest national **AI datathon**; designed problems and rubric, and supervised judging.
 - Collaborated with academic & industry mentors to ensure high-quality competition standards.

SELECTED PROJECTS

- **CareSync** [GitHub](#) [Demo](#) 2025
Healthcare service platform for appointment booking and clinical workflow management. **Frontend: React** | **Backend: Spring Boot**.
- **Real Time Incident Reporting App** [GitHub](#) [Demo](#) 2025
Lightweight information-management application demonstrating database-backed operations and API-driven workflows. **Frontend: Flutter** | **Backend: FastAPI**.
- **Criticup** [GitHub](#) [Demo](#) 2024
Database-driven web application emphasizing relational schema design, SQL queries, and CRUD operations. **Frontend: HTML, CSS, Bootstrap** | **Backend: Django**.
- **Side-Channel Attack** [GitHub](#) 2025
Implemented and analyzed a side-channel attack pipeline as part of a security course assignment, focusing on threat modeling, leakage analysis, and reproducible experimentation.
- **Slowloris Attack Project** [GitHub](#) [Demo](#) 2025
Controlled educational implementation of a Slowloris-style DoS attack to study server resource exhaustion behavior and defensive considerations.

ACADEMIC COURSEWORK & MATERIALS

- **BUET CSE Academic Compendium** [GitHub](#)

2022 – Present

Comprehensive collection of undergraduate computer science course materials and assignments at BUET. Includes structured solutions, implementations, and documentation covering core areas such as Data Structures, Algorithms, Databases, Operating Systems, Computer Networks, etc. .

TECHNICAL SKILLS

- **Programming:** Python, PyTorch, TensorFlow, Keras, C/C++, Java, SQL, JavaScript, Dart
- **ML/AI:** Computer Vision, Deep Learning, NLP, Unsupervised Learning, Agent-Based Modeling
- **Specialized:** Cryo-ET Processing, Medical Imaging, Bioimage Analysis, Mesa, Flutter
- **Development:** Django, HTML/CSS, Node, React, Spring, Database Design, Git, $\LaTeX$

REFERENCES

1. **Dr. Mashfiqui Rabbi** — Research Assistant Professor, Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign.
Email: mrabbi@illinois.edu | [Faculty Profile](#)
2. **Prof. Dr. James M. Rehg** — Founder Professor, Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign.
Email: jrehg@illinois.edu | [Faculty Profile](#)
3. **Dr. Min Xu** — Associate Professor, Department of Computational Biology, Carnegie Mellon University.
[Faculty Profile](#)
4. **Dr. Ben Lengerich** — Assistant Professor, University of Wisconsin–Madison.
Email: lengerich@wisc.edu
5. **Prof. Dr. M. Sohel Rahman** — Professor, Department of CSE, BUET.
Email: sohel.kcl@gmail.com | [Google Scholar](#)
6. **Prof. Dr. Md. Shamsuzzoha Bayzid** — Professor, Department of CSE, BUET.
Email: shamsbayzid@gmail.com | [Google Scholar](#)